

Calc 1: Related Rates & Review

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DIRECTIONS

Problems 1-6: Related Rates — draw a diagram, write an equation, differentiate with respect to time.

Problems 7-9: Mean Value Theorem — find c in (a,b) where $f'(c) = [f(b)-f(a)]/(b-a)$. Problem 10: find k so the function is continuous (factor and simplify the limit).

1 Related Rates (MC):

Conical reservoir: depth 20ft, top radius 10ft. Surface falls at $\frac{1}{2}$ ft/hr. Rate water leaves (ft³/hr) when depth = 8ft?

Answer: _____

2 Related Rates (MC):

Box: $h=10$ in, length grows 2in/s, width shrinks 4in/s. Rate of volume change when $l=8$ in and $w=6$ in?

Answer: _____

3 Related Rates:

A 20ft ladder leans against a house. Its foot is 12ft from the house and moving away at 2ft/s. How fast is the top sliding down?

Answer: _____

4 Related Rates:

Conical paper cup: diameter = height = 4in. Water leaks at $\frac{1}{2}$ in³/s. Rate the level drops when the surface diameter = 2in?

Answer: _____

5 Related Rates:

Car A goes east 40mph, Car B goes south 60mph (90mi from intersection at noon). At what rate is the distance between them changing at 1pm?

Answer: _____

6 Related Rates (3 parts):

For $V = \pi r^2 h$ (cylinder), find dV/dt in terms of: (a) dh/dt when r is constant (b) dr/dt when h is constant (c) both dh/dt and dr/dt when neither is constant

Answer: _____

7 Find c (MVT) on $[1, 4]$:

$$f(x) = x^2 - 5x + 4$$

Answer: _____

8 Find c (MVT) on $[1, 3]$:

$$f(x) = 3x^2 + 4x - 3$$

Answer: _____

9 Find c (MVT) on $[-2, 4]$:

$$f(x) = x^3 - 1$$

Answer: _____

10 Find k (Continuity at $x = 3$):

$$f(x) = \frac{x^2 - 2x - 3}{x - 3}, \quad x \neq 3$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

#1: $V = (\pi/3)r^2h$, $r = h/2$ so $V = (\pi/12)h^3$; $dV/dt = (\pi/4)h^2 \cdot dh/dt$; at $h=8$: 8π . #3: $x^2 + y^2 = 400$; $2x(dx/dt) + 2y(dy/dt) = 0$; $y=16$, $dy/dt = -3/2$. #4: $r = h/2$ (similar triangles); $V = (\pi/12)h^3$; at $h=2$: $dh/dt = 1/(2\pi)$. #5: at 1pm $A=40\text{mi}$, $B=30\text{mi}$, $D=50\text{mi}$; $dD/dt = -4\text{mph}$. #6 full answer: (a) $\pi r^2 \cdot dh/dt$, (b) $2\pi r \cdot h \cdot dr/dt$, (c) both combined.

1 Related Rates (MC):

Conical reservoir: depth 20ft, top radius 10ft. Surface falls at $\frac{1}{2}$ ft/hr...

$$= 8\pi \text{ ft}^3/\text{hr} \text{ (Choice B)}$$

2 Related Rates (MC):

Box: $h=10\text{in}$, length grows 2in/s , width shrinks 4in/s . Rate of volume cha...

$$= -200 \text{ in}^3/\text{s} \text{ (Choice D)}$$

3 Related Rates:

A 20ft ladder leans against a house. Its foot is 12ft from the house and...

$$= \frac{3}{2} \text{ ft/s}$$

4 Related Rates:

Conical paper cup: diameter = height = 4in. Water leaks at $\frac{1}{2} \text{ in}^3/\text{s}$. Rate...

$$= \frac{1}{2\pi} \text{ in/s}$$

5 Related Rates:

Car A goes east 40mph, Car B goes south 60mph (90mi from intersection at...

$$= -4 \text{ mph}$$

6 Related Rates (3 parts):

For $V = \pi r^2 h$ (cylinder), find dV/dt in terms of: (a) dh/dt when r is con...

$$= \frac{dV}{dt} = \pi r^2 \frac{dh}{dt} + 2\pi r h \frac{dr}{dt} \text{ (c)}$$

7 Find c (MVT) on [1, 4]:

$$f(x) = x^2 - 5x + 4$$

$$= c = \frac{5}{2}$$

8 Find c (MVT) on [1, 3]:

$$f(x) = 3x^2 + 4x - 3$$

$$= c = 2$$

9 Find c (MVT) on [-2, 4]:

$$f(x) = x^3 - 1$$

$$= c = 2$$

10 Find k (Continuity at $x = 3$):

$$f(x) = \frac{x^2 - 2x - 3}{x - 3}, x \neq 3$$

$$= k = 4$$

